RootIQ — OV POC Business Requirements (v2)

AI Root Cause, Risk & Prevention for Overage (OV) Exceptions

Field	Value
Owner	Saia LTL Freight — OS&D Operations
Phase	POC Phase 1 — **OV only**
Basis	v1 as-built BRD + target vision (root cause → pattern → risk → prevention → measure)
Status	Draft for review

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1. Objective

Show that AI can do more than explain why overages happened. For OV, the POC must demonstrate the full loop:

Exception → Root cause → Pattern → Risk → Prevention action → Measured result

Success means an OS&D leader can see *what* causes OVs, *where and when* they concentrate, *which shipments are at risk before it happens*, *what action to take*, and *whether the action reduced the OV rate*.

2. Scope

In scope (Phase 1): OV exceptions only; 12-category root-cause taxonomy; pattern mining; explainable OV risk score; simulated scan-gap alert; recommendation and action tracking; rate-based effectiveness measurement; KPI dashboard.

Deferred (Phase 2+): NB, AS, OH, MSD, and shortage / damage / refusal types (DS, BNF, PR, RD, RF, DD); live AS400 / OD400 / manifest integration; multi-app delivery of recommendations; semantic deduplication; authentication and RBAC; trained ML risk model.

The taxonomy, pipeline and API are designed type-agnostic so Phase 2 adds exception types, not rebuilds.

3. Users

Role	Needs
OS&D clerk	Log OV, investigate, clear; get live cause suggestions
OS&D analyst	Review and override AI results; work the review queue
Dock worker	Staging quantity check; see risk before handling
Terminal supervisor	Watch list, risk alerts, attention flags, assigned actions
Operations leadership	KPIs, patterns, recommendation status, measured impact

4. What changes from v1

Area	Keep	Fix	Add

Capture	Log form, six checks, PRO validation	Add *found-at terminal*, *handler*; fix value-range boundaries	Shipment volume base, cost fields
Classification	Taxonomy, cache, audit history, override	Override always wins; fallbacks never cached; review queue	—
Patterns	Terminal, commodity, window	Define "terminal" precisely	Shipper, lane, hour/shift, checkpoint
Prevention	Watch list, Notes Assist, staging check, flags	Scans run on schedule (or label as on-demand)	Risk score, scan-gap alert
Recommendations	Evidence IDs, escalation	—	Owner, workflow, confidence, expected impact
Measurement	Before/after, lineage	Use **rates**, not counts	Comparison group, minimum sample
Reporting	Distribution, cache, accuracy	—	KPI cards, cost avoided

5. Functional requirements

A. Capture and data foundations

- **FR-01** Capture the **found-at terminal** (where the freight was discovered) separately from shipment origin and destination. All "terminal" analytics use found-at unless stated.
- **FR-02** Define **handler** as a single field (assigned-to at time of discovery). Handler analytics use this field only.
- **FR-03** Value ranges use non-overlapping bounds: 1 = $0–249.99, 2 = $250–1,000, 3 = > $1,000. `dollar_amount` is required for range 3.
- **FR-04** Load a **shipment volume base**: shipment and piece counts per period by shipper, lane, origin terminal, commodity and package type. Needed for every rate. Synthetic if real data is unavailable, and labelled as such.
- **FR-05** Load a **scan-event feed** (PRO, checkpoint, terminal, timestamp). Synthetic for the POC, with a defined interface so AS400 can replace it later.
- **FR-06** Compute `content_hash` at classification time from the current notes, not at creation.

B. Root-cause classification

- **FR-07** Classify OV notes into the 12 categories plus the 7 contributing factors, with reasoning and confidence. The AI also receives `additional_details`, and `additional_details` is included in the dedup key.
- **FR-08** Classification is triggered when the investigation is marked complete, with analyst re-run available. Edited notes produce a new hash and a new appended row.
- **FR-09** `needs_review = true` when confidence is below a configured threshold (default 0.6) or on

fallback. A **Review queue** screen lists these.

- **FR-10** AI failure returns `Other / 0.0 / needs_review`, flagged `failure = true`. Failure results are **never cached** and are excluded from the category distribution.
- **FR-11** Human override is appended with user and mandatory reason. The effective classification is the latest override if one exists, otherwise the latest AI row. A cache hit never supersedes an override.
- **FR-12** Define and document the overlapping categories (Documentation Errors vs Labeling Error; Improper Packaging vs Handling Damage) with a tie-break rule in the prompt.

C. Pattern mining

- **FR-13** Aggregate effective classifications by: category × found-at terminal, shipper, lane, commodity, package type, hour/shift, and scan checkpoint, over selectable windows.
- **FR-14** Report each pattern as **rate** (OVs per 1,000 shipments) against the network baseline, with count and share. Flag patterns above baseline by a configurable margin (default +25%) and a minimum event count (default 10).
- **FR-15** Each pattern lists its supporting exception IDs.

D. OV risk score (proactive)

- **FR-16** Score a shipment 0–100 at two touchpoints: **receiving** and **staging/loading**.
- **FR-17** Method: smoothed historical OV rate for the shipment's shipper, lane, origin terminal, commodity, package type and shrink-wrap status, compared to baseline and converted to a percentile. Rules-based and explainable. No trained model in Phase 1.
- **FR-18** Every score shows its **top contributing factors** and the historical evidence behind each.
- **FR-19** Bands: Low (<50), Elevated (50–74), High (≥75), with a recommended preventive action per dominant root cause.
- **FR-20** Risk is **not** shown on the exception log form (retained from v1 BR-12).

E. Scan-gap alert (simulated)

- **FR-21** Using the scan-event feed, raise an alert when an expected checkpoint scan is missing beyond the normal processing window for that lane. The alert names the shipment, missing checkpoint and elapsed time.
- **FR-22** Alerts are labelled "simulated data" in the POC.

F. Recommendations and actions

- **FR-23** Generate 3–5 recommendations per scope from the aggregated pattern table. Each stores category, scope, pattern addressed, **confidence**, **expected impact** (range, labelled as estimate), supporting exception IDs, and date issued.
- **FR-24** Supporting IDs are attached by system query, not by the model. The UI states this.
- **FR-25** Each recommendation has an **owner** (terminal or role) and the workflow `proposed → assigned → implemented → monitoring → effective | ineffective`. Terminal states for replaced items are `superseded`, and `ineffective` is kept only on the item that triggered the replacement.
- **FR-26** Effectiveness is measured from the **implemented** date, not the issued date.
- **FR-27** Escalation (v1 rule retained): an ineffective recommendation generates a next-level replacement. At the ceiling (default 2) an attention flag is raised instead.

G. Measurement

- **FR-28** Effectiveness metric: OV rate per 1,000 shipments in the affected scope, before vs after implementation, using equal-length windows (default 4 weeks).
- **FR-29** Require a minimum sample (default 30 OVs across both windows) before judging. Otherwise show "insufficient data".
- **FR-30** Where possible, compare against a **control group** (similar scopes without the action) and show the difference. Results are worded "associated with a X% change", never "caused".
- **FR-31** Prevention checks resolve to `prevented` or `recurred` by this rule: a `mismatch_corrected` check is `prevented` if no OV arises for that PRO within 14 days, else `recurred`. `match` results are recorded but not counted as preventions. `mismatch_confirmed` raises a review item.

H. Dashboard and KPIs

- **FR-32** KPI cards: total OVs (vs prior period), **OV rate**, high-risk shipments, preventive actions taken, OVs prevented (from FR-31), **estimated cost avoided**.
- **FR-33** Cost avoided = prevented count × average OV handling cost (configurable) + dollar amounts where captured. Shown as an estimate with its formula visible.
- **FR-34** Retain: root-cause mix, terminal and commodity breakdowns, cache statistics, accuracy, evidence drawer.
- **FR-35 Prevention Opportunities** list ranked by excess OV volume over baseline, each linking to its recommendation and owner.

I. Point-of-work prevention (retained)

- **FR-36** Watch list (pre-handling), Notes Assist (advisory, TF-IDF), staging quantity check, and attention flags remain. Attention scans and effectiveness evaluation run on a **schedule** (daily, configurable) with a manual trigger.
- **FR-37** If no expected-shipment record exists for a staging check, the result is `no_reference` and is logged without blocking the worker.

6. Business rules

Retain v1 BR-01 to BR-15, with these amendments and additions:

ID	Rule
BR-04a	History is append-only. An override always takes precedence over later AI or cache rows.
BR-05a	A cache hit makes no model call. Failure results are never cached.
BR-16	All effectiveness and risk metrics use rates against a shipment volume base, never raw counts alone.
BR-17	Impact figures are estimates and are labelled as such.
BR-18	Effectiveness is measured from implementation, not issuance.
BR-19	Overrides, acknowledgements and resolutions record user and reason.
BR-20	Synthetic data is visibly labelled wherever it drives a screen.

7. Non-functional requirements

- **Resilience:** fully usable when the model is down.
- **Determinism:** `temperature = 0`, JSON-constrained output.
- **Privacy:** real exports are anonymised (shipper/consignee names, contact details) before loading. Notes are not sent to a hosted model until approved by Saia security. Local model is the default.
- **Auditability:** every AI result, override, flag action and recommendation transition stores actor, reason, model and timestamp.
- **Handler analytics:** shown to supervisors and leadership only, framed as process signals, not performance ratings.
- **Testability:** test suite runs with the model mocked.

8. Success criteria (proposed targets, to confirm)

Metric	Target	Measured by
Classification accuracy	≥ 85% agreement on a held-out labelled set of ≥ 100 OVs, **excluding** the few-shot examples	Accuracy report
Review workload	≤ 20% of classifications flagged `needs_review`	Review queue
Cache	≥ 90% of re-runs of unchanged notes served without a model call	Cache stats
Risk score validity	Top-decile scored shipments show ≥ 3× the baseline OV rate on historical back-test	Back-test report
Pattern quality	Analysts rate ≥ 80% of top-10 patterns as "real and actionable"	Analyst review
Traceability	100% of recommendations list supporting exceptions	Evidence drawer
Measurability	Every implemented recommendation reaches a verdict or "insufficient data"	Effectiveness report
Resilience	Zero user-facing failures with the model down	Fallback tests

9. Assumptions, dependencies, risks

Assumptions: an anonymised OV export with at least 6 months of history is available; shipment volume data can be supplied or synthesised.

Dependencies: Saia approval for data use; a named owner per terminal for action tracking; agreement on average OV handling cost.

Risk	Mitigation
No real volume or scan data	Synthetic base, clearly labelled; define the real feed interface now
Risk score cannot be proven without ground truth	Historical back-test (FR success criteria)
Before/after confounded by volume and seasonality	Rates, control group, minimum sample

Before/after confounded by volume and seasonality	Rates, control group, minimum sample, "associated with" wording
AI classifications feed Notes Assist and patterns (feedback loop of errors)	Review queue; override precedence; accuracy tracking
Handler-level data misused	Restricted audience, process framing, RBAC in Phase 2

10. Open questions

1. Do real scan events, shipment volumes and handling costs exist, and who owns them?
2. Is the notes data cleared to go to a hosted model?
3. What accuracy and risk-lift targets does Saia consider a pass?
4. Who owns recommendation actions at each terminal?
5. Is Phase 2 priority shortages (AS, DS, BNF) or damages?

11. Roadmap

Phase	Content
1 (this POC)	OV loop end to end, as above
2	NB, AS, OH, MSD and shortage/damage/refusal types; AS400 and manifest integration; RBAC and authentication
3	Trained risk model, semantic dedup, multi-app delivery of recommendations, scheduled automation at scale